Screening task for Autumn Internship under ANN Modeling of Binary Azeotropic VLE

Task: ANN Surrogate Modeling for Binary VLE

Build an **Artificial Neural Network (ANN)** model in **Python (preferred)** or **Scilab** to predict **vapor composition** in a **binary azeotropic system**.
The ANN should be trained on experimental or simulated VLE data and tested on its ability to **capture azeotropic behavior**.

Technical Requirements:

1. Basic knowledge of VLE thermodynamics (Raoult's Law, azeotropy, activity coefficient models).
2. Basic knowledge of Python/Scilab programming.
3. Familiarity with machine learning libraries (TensorFlow, PyTorch, or Scikit-learn).

Procedure to Submit:

1. **Dataset Preparation**
   - Choose a binary azeotropic system (e.g., Ethanol–Water, Acetone–Chloroform).
   - Collect/generate at least **300–500 equilibrium data points**: $(x_1,T,P)\rightarrow(y_1)$
   - Ensure dense sampling near the azeotropic composition.
2. **Model Development**
   - Preprocess data (normalize, split into train/validation/test).
   - Train an ANN with inputs (x_1,T,P) and output y_1.
   - Enforce mole fraction bounds (e.g., sigmoid activation).
3. **Evaluation**
   - Plot **parity plots**: $y_{1,exp}$ vs $y_{1,ANN}$
   - Detect azeotrope by solving $y_1 \approx x_1$ from the ANN predictions.
   - Compare predicted azeotropic composition & temperature with reference data.
   - Compare performance with Raoult's Law as a baseline.
4. **Submission Files**
   - ANN model script in **.py** or **.sce** format with clear comments.
   - A short **report (2–3 pages, PDF)** describing dataset, ANN design, training, and results.
   - Archive everything into a **.zip** file named "ANN_BinaryVLE.zip".

- Upload the zip file in your google drive and ensure to give view access to the uploaded zip file
- Submit the gdrive file link here.

Evaluation Criteria:

- **Complexity:** ANN design (baseline vs improved physics-informed approaches).
- **Accuracy:** Prediction error (MAE, RMSE) and correct azeotrope detection.
- **Presentation:** Quality of report, clarity of code, and visualization of results.